

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(a)(i) | <ol style="list-style-type: none"> <li>1 rate of oxygen produced / rate of photosynthesis, is low / stays constant<br/><b>or</b><br/>small decrease in, rate of oxygen produced / rate of photosynthesis ;</li> <li>2 light <u>intensity</u> was the limiting factor / AW ;</li> <li>3 (so) less / little, light energy absorbed / photolysis (of water) / photoactivation (of chlorophyll)/ light-dependent reactions / photophosphorylation ; <b>R</b> no light energy absorbed</li> </ol> | <b>3</b> |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

| Question         | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks            |                                                             |    |          |    |      |    |                            |   |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------|----|----------|----|------|----|----------------------------|---|
| 7(a)(ii)         | <p>any <b>three</b> from:</p> <p>1 rate of oxygen produced / rate of photosynthesis, decreased ;</p> <p>2 two paired data quotes with units ;</p> <table><tr><td>temperature / °C</td><td>rate of oxygen production / mm<sup>3</sup> h<sup>-1</sup></td></tr><tr><td>30</td><td>4 / 4.00</td></tr><tr><td>40</td><td>2.55</td></tr><tr><td>50</td><td>0.65 (<b>A</b> 0.62-0.65)</td></tr></table> <p>3 because, enzyme / ATP synthase / NADP reductase / oxygen evolving complex, <u>denatured</u> ; <b>A</b> rubisco</p> <p>4 change in active site / fewer ESCs ;</p> | temperature / °C | rate of oxygen production / mm <sup>3</sup> h <sup>-1</sup> | 30 | 4 / 4.00 | 40 | 2.55 | 50 | 0.65 ( <b>A</b> 0.62-0.65) | 3 |
| temperature / °C | rate of oxygen production / mm <sup>3</sup> h <sup>-1</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                  |                                                             |    |          |    |      |    |                            |   |
| 30               | 4 / 4.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                  |                                                             |    |          |    |      |    |                            |   |
| 40               | 2.55                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                                                             |    |          |    |      |    |                            |   |
| 50               | 0.65 ( <b>A</b> 0.62-0.65)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |                                                             |    |          |    |      |    |                            |   |
| 7(b)             | <p>any <b>three</b> from:</p> <p>1 light absorbed by, photosystem II / P680<br/><b>or</b><br/>photoactivation / photoionisation (of chlorophyll) ;</p> <p>2 photolysis (of water) ;</p> <p>3 water-splitting enzyme / oxygen evolving complex ;</p> <p>4 water splits into protons, electrons and oxygen ; <b>A</b> equation</p>                                                                                                                                                                                                                                        | 3                |                                                             |    |          |    |      |    |                            |   |